



MATH-619

A CNN-LSTM Hybrid Model for Multivariate Stock Price Forecasting Using Technical and Sentiment Indicators

Background and Review of the Literature

Student Name & ID ABDULLAH ALHARBI g202415480	Advisor Dr. Mousa Ahmad Al-Bashrawi Co-advisor Mr. Mohammed Agbawi
--	---

1) Introduction to Stock Price Prediction Literature

Stock market prediction has been an active area of research for decades, with significant progress made through the evolution from traditional statistical methods to modern deep learning approaches. The field has witnessed a transformative shift as machine learning techniques have demonstrated remarkable capabilities in capturing complex, nonlinear patterns inherent in financial time series data. According to Desai (2023), writing in the *Harvard Business Review*, the finance industry has been transformed in the last decade by the same forces driving artificial intelligence: the diffusion of ever more powerful computing and the profusion of data. This transformation suggests that AI will significantly impact certain industries, particularly benefiting larger players with access to substantial data resources.

The challenge of stock price prediction stems from the inherently noisy, chaotic, and highly nonlinear nature of financial markets. A comprehensive systematic review by Sezer et al. (2020) established that deep learning models, particularly recurrent neural networks and their variants, have emerged as state-of-the-art approaches for financial time series forecasting. Their analysis of research from 2005-2019 revealed that LSTM-based architectures are the most commonly adopted, with applications spanning stock forecasting, index prediction, trend analysis, and volatility modeling. The McKinsey Global Institute has reported that 60% of financial institutions are now using AI and machine learning technologies to enhance decision-making, particularly in risk management, fraud detection, and investment strategies, with machine learning models improving the accuracy of financial predictions by 20-30% compared to traditional methods (McKinsey & Company, 2020).

2) Deep Learning Architectures for Stock Prediction

2.1 LSTM Networks in Financial Forecasting

Long Short-Term Memory networks have emerged as the dominant approach for financial time series prediction. Fischer and Krauss (2018), publishing in the *European Journal of Operational Research*, conducted a seminal study demonstrating that LSTM networks significantly outperform traditional machine learning methods including Random Forests and logistic regression in predicting stock market movements. Their research on S&P 500 constituents showed that LSTM models achieved daily returns of 0.46% compared to 0.27% for Random Forests, highlighting the practical economic significance of deep learning in

financial markets. The authors emphasized that LSTM's ability to capture long-term dependencies in sequential data makes it particularly suited for modeling financial time series where past patterns influence future movements.

A recent study published in *Humanities and Social Sciences Communications* by Lamine et al. (2023) applied LSTM models to predict financial time series during the COVID-19 pandemic crisis, demonstrating that the architecture maintains robust predictive performance even during periods of extreme market volatility. Their findings indicated that LSTM networks outperformed ARIMA-type models across various asset classes including equities, bonds, and commodities, achieving consistently lower RMSE and MAE values. This practical validation under crisis conditions underscores the model's applicability for real-world forecasting scenarios where market regime changes occur unexpectedly.

2.2 CNN-LSTM Hybrid Architectures

The combination of CNNs and LSTMs represents a significant advancement in hybrid deep learning architectures for stock prediction. Lu et al. (2020), in a highly-cited paper published in *Complexity*, proposed a foundational CNN-LSTM framework where CNNs extract effective features from multivariate input data while LSTMs model the temporal dependencies. Their empirical results demonstrated that the hybrid approach combines the advantages of CNNs (extracting spatial features from technical indicators) with LSTMs (capturing time-series dependencies), achieving the highest prediction accuracy across multiple stock datasets. The study established that this complementary architecture creates a more comprehensive learning framework than standalone models.

Vijh et al. (2022), publishing in *Intelligent Systems with Applications*, provided a comprehensive review of multiple hybrid deep learning approaches for stock prediction. Their analysis emphasized that hybrid CNN-LSTM models can forecast either accurate stock rates or general trends with dynamic capture of swift market changes, making them particularly suitable for both short-term trading and longer-term investment strategies. The review noted that these hybrid architectures consistently demonstrate lower MSE, RMSE, and MAE values compared to single-model approaches, with accuracy improvements ranging from 5-15% depending on the market and time period analyzed. The authors concluded that combining different deep learning paradigms addresses individual model limitations while leveraging their respective strengths.

3) Technical Indicators in Predictive Models

Technical indicators serve as crucial input features for machine learning models, capturing various aspects of price dynamics including trend, momentum, and volatility. Dhokane and Agarwal (2024), publishing in the *International Journal of Intelligent Systems and Applications in Engineering*, conducted a comprehensive study using LSTM networks with Bollinger Bands, RSI, and MACD indicators as features for predicting stocks on the National Stock Exchange of India. Their research demonstrated that the combination of these technical indicators significantly enhances model performance compared to using raw price data alone. The Bollinger Bands captured volatility patterns, RSI provided overbought and oversold signals, while MACD indicated momentum shifts and trend reversals. The study achieved improved prediction accuracy when these indicators were combined with OHLC price data.

Research from the practitioner community has further validated the importance of technical indicator integration. A study published in *Journal of Risk and Financial Management* by Sangeetha and Alfia (2024) demonstrated that hybrid approaches combining technical indicators with deep learning models provide a more holistic perspective on stock market trends. Their LSTM-CNN hybrid model evaluated using RMSE, MAE, MAPE, and R^2 metrics on the S&P 500 index over a 14-year period showed superior predictive performance compared to traditional models including SVMs, Random Forest, and ARIMA. The study emphasized that EMA-20 and EMA-50 crossovers, when combined with momentum indicators like RSI and MACD, create robust feature sets that capture both short-term fluctuations and longer-term trend patterns.

4) VIX as a Market Sentiment Proxy

The CBOE Volatility Index (VIX), often referred to as the 'fear index,' has emerged as a significant predictor variable in stock market forecasting models. Bekaert and Hoerova (2014), in a foundational paper published in the *Journal of Econometrics*, established the theoretical basis for incorporating VIX into predictive frameworks by demonstrating strong correlations between VIX movements and subsequent market returns, particularly during periods of market stress. Their research provided evidence that VIX contains forward-looking information about market uncertainty beyond what is captured by historical volatility measures.

Recent machine learning applications have further explored VIX predictability. Zhang et al. (2024), publishing in *Quantitative Finance*, investigated VIX forecasting using adaptive machine learning techniques, establishing that daily VIX can be predicted with higher accuracy than previously documented. Their analysis demonstrated that machine learning approaches, particularly RNN and LSTM architectures, provide improved forecasts over conventional methods including linear regression and ARIMA models. A complementary study by Li et al. (2024) proposed hybrid LSTM-GARCH models that incorporate VIX as an input feature for S&P 500 volatility forecasting. Their comprehensive analysis spanning 2000-2023 demonstrated that models including VIX inputs achieve significantly higher predictive accuracy compared to technical-only models, finding that VIX provides crucial forward-looking information about market uncertainty.

5) Sentiment Analysis in Stock Prediction

The integration of sentiment analysis into stock prediction models represents a significant research frontier with strong practitioner interest. Bollen et al. (2011), in a landmark paper published in the *Journal of Computational Science*, demonstrated that Twitter mood predicts the stock market with approximately 87% accuracy in predicting directional changes in the Dow Jones Industrial Average. Using Self-Organizing Fuzzy Neural Networks with mood indicators extracted from tweets, their research established the foundational relationship between public sentiment expressed on social media and subsequent market movements. This study has been widely cited in both academic and practitioner literature as evidence that alternative data sources contain predictive signals not captured by traditional technical analysis.

Building on this foundation, Khan et al. (2022), publishing in *Expert Systems with Applications*, proposed a cooperative deep learning model combining autoencoders, LSTM, and sentiment analysis. Their DAE-LSTMSA and DAE-GRUSA architectures achieved superior performance across multiple stock indices, with sentiment features contributing 3-8% improvement in prediction accuracy. The multi-stage approach effectively denoises historical data while incorporating sentiment signals from news headlines. Similarly, Maqsood et al. (2020), publishing in the *International Journal of Information Management*, demonstrated that local and global event sentiment significantly impacts stock exchange forecasting, with deep learning methods effectively mining sentiment features from multiple

sources. Their research emphasized the importance of daily aggregation of sentiment scores and proper alignment with market trading days to maximize predictive value.

6) Explainable AI in Financial Forecasting

The deployment of deep learning models in financial contexts necessitates interpretability and transparency, particularly for practitioner adoption. Lundberg and Lee (2017), in their influential paper presented at *Advances in Neural Information Processing Systems*, introduced SHAP (SHapley Additive exPlanations), providing a unified approach to interpreting model predictions based on game-theoretic principles. The framework has been widely adopted in financial forecasting to quantify feature contributions and build stakeholder confidence in model outputs. This addresses the 'black box' concern that has historically limited deep learning adoption in finance.

Kumar et al. (2025), publishing in *Annals of Operations Research*, demonstrated the significance of predictors in stock return predictions using explainable AI techniques. Their research showed that SHAP analysis can reveal the relative importance of technical indicators versus sentiment features, enabling informed feature selection and model refinement. A comprehensive study published in *Heliyon* by Raza et al. (2024) extended explainable AI applications to enhance investor confidence through interpretable model outputs. Their framework demonstrated that combining SHAP and LIME (Local Interpretable Model-agnostic Explanations) provides both global feature importance rankings and local prediction explanations, achieving an average accuracy of 94.9% while maintaining interpretability.

7) Industry Perspectives on AI in Finance

Practitioner-oriented sources provide valuable perspectives on AI implementation in financial markets. Iansiti and Lakhani (2020), writing in the *Harvard Business Review*, described how AI-driven firms operate fundamentally differently from traditional companies, with algorithms at the core of value delivery. Their analysis of companies like Ant Financial illustrated how AI-based financial services can serve more than 10 times as many customers as traditional banks with less than one-tenth the employees. The authors emphasized that the future competitive advantage lies in firms' ability to scale AI-based processes, which never stop improving with more data.

McKinsey & Company (2020) provided practical guidance on applying machine learning in capital markets for pricing, valuation adjustments, and market risk. Their research highlighted that during the COVID-19 pandemic volatility, traditional financial models faced significant challenges, prompting leading institutions to incorporate advanced machine learning techniques including neural networks. The report emphasized that ML-enhanced models can provide more accurate and timely valuations, with speed being essential in unpredictable market environments. According to the MIT Sloan Management Review (2020), only 10% of organizations achieve significant financial benefits with AI, but those that develop multiple effective ways for humans and AI to learn together are six times more likely to amplify their success with AI implementation.

8) Summary and Research Gap

This literature review reveals substantial progress in applying deep learning techniques to stock market prediction across both academic research and practitioner applications. Key findings include: (1) Hybrid CNN-LSTM architectures consistently outperform standalone models by capturing both spatial patterns in technical indicators and temporal dependencies in price sequences; (2) Technical indicators such as RSI, MACD, EMA, and Bollinger Bands provide meaningful features that improve prediction accuracy by 5-15%; (3) VIX serves as an effective proxy for market-wide sentiment and uncertainty, providing forward-looking information; (4) Social media sentiment, particularly from Twitter, demonstrates predictive value for stock movements with reported accuracy improvements of 3-8%; and (5) Explainable AI techniques enhance model transparency without sacrificing performance, addressing a key barrier to practitioner adoption.

Despite these advances, gaps remain in the literature. Few studies comprehensively integrate technical indicators, market volatility measures, and pre-computed sentiment scores within a unified deep learning framework. Additionally, most research focuses on single-stock or index prediction rather than multi-company portfolios with sector-specific characteristics. The proposed CNN-LSTM model addresses these gaps by integrating TradingView technical indicators (RSI, MACD, EMA, Bollinger Bands), VIX data, and Kaggle Twitter sentiment scores for technology sector companies (AAPL, GOOG, AMZN, MSFT, TSLA), providing both technical-only and sentiment-enhanced model configurations for systematic comparison with explainable AI techniques for transparency.

References

- Bekaert, G., & Hoerova, M. (2014). The VIX, the variance premium and stock market volatility. *Journal of Econometrics*, 183(2), 181–192. <https://doi.org/10.1016/j.jeconom.2014.05.008>
- Bollen, J., Mao, H., & Zeng, X. (2011). Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1), 1–8. <https://doi.org/10.1016/j.jocs.2010.12.007>
- Desai, M. A. (2023, August 9). What the finance industry tells us about the future of AI. *Harvard Business Review*. <https://hbr.org/2023/08/what-the-finance-industry-tells-us-about-the-future-of-ai>
- Dhokane, R. M., & Agarwal, S. (2024). LSTM deep learning based stock price prediction with Bollinger Band, RSI, MACD, and OHLC features. *International Journal of Intelligent Systems and Applications in Engineering*, 12(3), 475–484.
- Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669. <https://doi.org/10.1016/j.ejor.2017.11.054>
- Iansiti, M., & Lakhani, K. R. (2020, January–February). Competing in the age of AI. *Harvard Business Review*. <https://hbr.org/2020/01/competing-in-the-age-of-ai>
- Khan, W., Malik, U., Ghazanfar, M. A., Azam, M. A., Alyoubi, K. H., & Alfakeeh, A. S. (2022). A cooperative deep learning model for stock market prediction using deep autoencoder and sentiment analysis. *Expert Systems with Applications*, 203, 117426. <https://doi.org/10.1016/j.eswa.2022.117426>
- Kumar, A., Das, S. R., & Chattopadhyay, M. (2025). Significance of predictors: Revisiting stock return predictions using explainable AI. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-025-06717-2>
- Lamine, A., Adjout, F., & Tarfi, H. (2023). Financial time series prediction under COVID-19 pandemic crisis with Long Short-Term Memory (LSTM) network. *Humanities and Social Sciences Communications*, 10, 522. <https://doi.org/10.1057/s41599-023-02042-w>
- Li, Y., Chen, X., & Wang, S. (2024). The hybrid forecast of S&P 500 volatility ensembled from VIX, GARCH and LSTM models. *arXiv*. <https://doi.org/10.48550/arXiv.2407.16780>
- Lu, W., Li, J., Li, Y., Sun, A., & Wang, J. (2020). A CNN-LSTM-based model to forecast stock prices. *Complexity*, 2020, Article 6622927. <https://doi.org/10.1155/2020/6622927>
- Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* 30 (pp. 4765–4774). Curran Associates, Inc.
- Maqsood, H., Mehmood, I., Maqsood, M., Yasir, M., Afzal, S., Aadil, F., Selim, M. M., & Muhammad, K. (2020). A local and global event sentiment based efficient stock exchange forecasting using deep learning. *International Journal of Information Management*, 50, 432–451. <https://doi.org/10.1016/j.ijinfomgt.2019.07.011>
- McKinsey & Company. (2020, October 29). Applying machine learning in capital markets: Pricing, valuation adjustments, and market risk. *McKinsey Insights*.

<https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/applying-machine-learning-in-capital-markets>

MIT Sloan Management Review. (2020, October 20). Annual MIT Sloan Management Review—Boston Consulting Group study finds significant financial benefits with AI. *MIT Sloan Management Review*. <https://sloanreview.mit.edu>

Raza, A., Bukhari, S., Asghar, M. Z., Bangash, J. I., & Hussain, A. (2024). An explainable deep learning approach for stock market trend prediction. *Heliyon*, 10(22), e40095. <https://doi.org/10.1016/j.heliyon.2024.e40095>

Sangeetha, K., & Alfia, S. (2024). Hybrid machine learning models for long-term stock market forecasting: Integrating technical indicators. *Journal of Risk and Financial Management*, 18(4), 201. <https://doi.org/10.3390/jrfm18040201>

Sezer, O. B., Gudelek, M. U., & Ozbayoglu, A. M. (2020). Financial time series forecasting with deep learning: A systematic literature review: 2005–2019. *Applied Soft Computing*, 90, 106181. <https://doi.org/10.1016/j.asoc.2020.106181>

Vijh, M., Chandola, D., Tikkiwal, V. A., & Kumar, A. (2022). A comprehensive review on multiple hybrid deep learning approaches for stock prediction. *Intelligent Systems with Applications*, 16, 200111. <https://doi.org/10.1016/j.iswa.2022.200111>

Zhang, L., Liu, W., & Chen, Y. (2024). Predicting VIX with adaptive machine learning. *Quantitative Finance*, 24(12), 1785–1802. <https://doi.org/10.1080/14697688.2024.2439458>